PH7002: Concepts in Statistical Mechanics
Assignment 3 Solutions

1 Sethna 9.14

We model the dissipation of energy into heat using Kelvin damping, with damping constant d^2:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} + d^2 \frac{\partial}{\partial t} \frac{\partial^2 u}{\partial x^2}$$

(a) We use the ansatz $u(x,t) = e^{i(kx - \Omega_k t)}$, where Ω_k denotes the complex frequency $\Omega_k = \omega_k - i\Gamma_k$. Note that $\dot{u} = -i\Omega u$, $\ddot{u} = -\Omega^2 u$, $u' = iku$, and $u'' = -k^2 u$. From the wave equation, we have $\Omega^2 + id^2k^2\Omega - c^2k^2 = 0$ whose roots are

$$\Omega_\pm = -\frac{i}{2} d^2 k^2 \pm \frac{k}{2}\sqrt{4c^2 - d^4k^2} \ .$$

We can therefore identify $\omega_\pm = \pm k\sqrt{4c^2 - d^4k^2}/2$ and the damping $\Gamma = d^2k^2/2$. In the long wavelength limit, k approaches zero and thus $\omega_\pm = \pm ck$. If we define the quality factor Q_k to be 2π times the number of periods of oscillation needed for the energy in the wave to decay by a factor of $1/e$, i.e. $Q_k = 2\pi N$ where $N = \widetilde{t}/T$ and T is the period of oscillation. We know that $E \propto |u(x,t)|^2 = e^{-2\Gamma t}$, we need $\widetilde{t} = (2\Gamma)^{-1}$ so that the energy in the wave to decay by a factor of $1/e$. Moreover, with $\omega = 2\pi/T$ the quality factor can be written as $Q_k = \omega/(2\Gamma)$. As $k \to 0$, it is easy to see that Q_k diverges as k^{-1}.

(b) The frequency ω vanishes when $\lambda = 2\pi/k = \pi d^2/c$.

(c) Since the energy falls of with a distance L as $e^{-2\Gamma L/c}$, and with the long wavelength assumption, i.e. $\Gamma \propto \omega^2$, if the frequency ω is damped by a factor of 3 the sound at 2ω will be damped by a factor of $3^4 = 81$.

2 Sethna 9.20

The free energy of superfluid particles is described by the Ginzburg–Landau theory

$$\mathcal{F} = \frac{1}{2} K |\nabla\psi|^2 + V(|\psi|) \ ,$$

where K is the superfluid stiffness per particle and $V(|\psi|) = \alpha|\psi|^2 + \beta|\psi|^4/2$. The local density is given by $\psi^\dagger(x)\psi(x)$ where $\psi = |\psi|e^{i\theta}$. In this question, we follow P. W. Anderson's derivation in Rev. Mod. Phys. **38**, 298 (1966) on the emergent superconducting equation of motions from the observation that particle number $|\psi|^2$ and ϕ are conjugate variables. The phase ϕ expresses the spontaneous-breaking of $U(1)$ gauge symmetry by superfluids, which through Noether's theorem would result in the conservation of particle number that we'll see in this question.

(a) To rewrite $\mathcal{F}$, we note that

$$\nabla\psi = (\nabla|\psi|)\, e^{i\phi} + i|\psi|e^{i\phi}\,(\nabla\phi) \ ,$$

and hence

$$|\nabla\psi|^2 = (\nabla|\psi|)^2 + |\psi|^2(\nabla\phi)^2 \ . \tag{1}$$

Plugging it into $\mathcal{F}$, we have

$$\mathcal{F} = \frac{1}{2}K(\nabla\phi)^2 + \text{terms independent of } \phi \ .$$

(b) We'll now minimize the functional $\mathcal{F}$ with respect to $\phi \to \phi + \delta\phi$, where $\delta\phi$ is the variation of ϕ.

$$\mathcal{F}[\phi + \delta\phi] = \mathcal{F}[\phi] + \frac{\partial\mathcal{F}}{\partial\phi}\delta\phi + \cdots \ .$$

Since the equation of motion is associated with the extremal of F, i.e. $\delta F = 0$, we perform integration on both sides by keeping only the term of order $\delta\phi$ to get

$$\int d^D x \frac{\partial\mathcal{F}}{\partial\phi}\delta\phi = \int d^D x \left(\mathcal{F}[\phi + \delta\phi] - \mathcal{F}[\phi]\right) \ .$$

Of course one can set $D = 1$ to work in a one-dimensional system, but we'll keep it general for now. Note that the term that is proportional to $(\nabla\phi)^2$ is the only term in $\mathcal{F}$ that is affected by the variation on ϕ. With $\nabla(\phi + \delta\phi) = \nabla\phi + \nabla(\delta\phi)$, we have

$$\int d^D x \frac{\partial\mathcal{F}}{\partial\phi}\delta\phi = \int d^D x \, \frac{K}{2}|\psi|^2\{(\nabla\phi + \nabla(\delta\phi))^2 - (\nabla\phi)^2\} = \int d^D x \, K|\psi|^2(\nabla\phi)(\nabla\delta\phi) \ .$$

Here, we've ignored the contribution that comes from $(\delta\phi)^2$.

(c) Following an integration by parts to the integral part (b), we have

$$\int d^D x \frac{\partial\mathcal{F}}{\partial\phi}\delta\phi = \left(K|\psi|^2\nabla(\phi)\delta\phi\Big|_{(D-1)\text{ boundary}} - \int d^D x \, \mathrm{div}(\nabla(\phi))\delta\phi \right) = -\int d^D x \, \mathrm{div}(K|\psi|^2\nabla(\phi))\delta\phi \ ,$$

where we've used the fact that the boundary term vanishes. From the equation above, it's easy to identify

$$\frac{\partial\mathcal{F}}{\partial\phi} = -\mathrm{div}(K|\psi|^2\nabla(\phi)) \ .$$

Now, we want to confirm that the first equation of motion is derived from $|\psi|^2$ and ϕ being conjugate variables. We already have the expression for supercurrent $\mathbf{J}$, and since

$$\psi^\dagger\nabla\psi = |\psi|\left(\nabla|\psi|\right) + i|\psi|^2\left(\nabla\phi\right) \ ,$$

the supercurrent can be written as

$$\mathbf{J} = \frac{K}{2i\hbar}\left(\psi^\dagger\nabla\psi - \psi\nabla\psi^\dagger\right) = \frac{K}{\hbar}\mathrm{Im}[\psi^\dagger\nabla\psi] = \frac{K}{\hbar}|\psi|^2\left(\nabla\phi\right) \ .$$

We can see that

$$\frac{\partial \mathcal{F}}{\partial \phi} = -\hbar \operatorname{div} \mathbf{J} \ .$$

Note that if $D = 1$, $\operatorname{div} \mathbf{J}$ is simply $\partial J/\partial x$. Since the charge conservation law states $\partial \rho/\partial t + \operatorname{div} \mathbf{J} = 0$, then

$$\frac{\partial \mathcal{F}}{\partial \phi} = \hbar \frac{\partial \rho}{\partial t} \ .$$

Therefore, the first equation of motion is indeed derived from formally treating $|\psi|^2$ and ϕ as conjugate variables.

Remarks: The second equation of motion is more subtle, it encodes the most noteworthy feature of a superfluid flow −a superfluid may undergo acceleration without frictional damping (i.e. zero viscosity)−below a certain temperature (The Nobel Prize in Physics 1962, 1972, 1978, 1996). This leads to some strange phenomena, for instance, see What is One Degree?. The second equation of motion was also used by Josephson to discuss the tunneling current between superconductors coupled through a thin barrier (The Nobel Prize in Physics 1973).

3 1D XY model

In this question, we analyze the thermodynamic properties of the one-dimensional XY model using transfer matrix formalism. The nearest-neighbor spin interactions of a XY model is described by

$$-\beta H = K \sum_{i=1}^{N-1} \mathbf{s}_i \cdot \mathbf{s}_{i+1} \ ,$$

with $\mathbf{s}_i = (\cos\theta_i, \sin\theta_i)$.

(a) The partition function reads

$$Z = \sum_{\{\mathbf{s}_1, \mathbf{s}_2 \ldots, \mathbf{s}_N\}} e^{\mathbf{s}_1 \cdot \mathbf{s}_2} e^{\mathbf{s}_2 \cdot \mathbf{s}_3} \cdots e^{\mathbf{s}_{N-1} \cdot \mathbf{s}_N} \ .$$

We then identify T as the transfer function with elements given by $T_{i,i'} = e^{\mathbf{s}_i \cdot \mathbf{s}_{i'}}$. Since $\mathbf{s}_i \cdot \mathbf{s}_{i'} = \cos\theta\cos\theta' + \sin\theta\sin\theta' = \cos(\theta - \theta')$ and $\{\theta_i\}$ being continuous, we can rewrite the elements of T in the basis of angles $\{\theta_i\}$ as

$$\langle \theta | T | \theta' \rangle = e^{K \cos(\theta - \theta')} \ .$$

(b) Here, we'll show that the transfer function can be diagonalized in the Fourier basis $\langle \theta | m \rangle \propto e^{im\theta}$ (up to a normalization constant). We have

$$\langle \theta | T | m \rangle = \langle \theta | T | \theta' \rangle \langle \theta' | m \rangle \propto \int_0^{2\pi} \frac{d\theta'}{2\pi} e^{K \cos(\theta - \theta') + im\theta'} = \left(\int_0^{2\pi} \frac{d\phi}{2\pi} e^{K \cos(\phi) + im\phi} \right) \langle \theta | m \rangle \ ,$$

where we've introduced $\phi = \theta - \theta'$. We can now identify the eigenvalues

$$\lambda_m = \int_0^{2\pi} \frac{d\phi}{2\pi} e^{K\cos(\phi)+im\phi} \ .$$

In its current form, the integral is tricky to solve because it won't be so straightforward for us to determine the right integration contour Γ due to the fact that the power of the exponent oscillates in the complex plane. However, note that at low temperature $K \gg 1$ and the angles $\{\theta_i\}$ vary slowly from site to site such that one can approximate $\cos\phi \approx 1 - \phi^2/2$. These two conditions allow us to remove the oscillating part and evaluate the integral in the complex plane easily via the saddle-point method. The integral now reads

$$\lambda_m \simeq e^K \int_0^{2\pi} \frac{d\phi}{2\pi} e^{-K\phi^2/2+im\phi} = \frac{e^K}{2\pi} \int_\Gamma dy\, e^{Kg(y)} \ ,$$

where $g(y) = -y^2/2 + imy/K$ and Γ is the integration contour in the y-plane. If we first take the Taylor expansion of $g(y)$ around $y_0 = im/K$, i.e. the point at which $g'(y) = 0$, and then transform $y \to y_0 + y/\sqrt{K}$, we get

$$\lambda_m \simeq \frac{e^{K+Kg(y_0)}}{2\pi\sqrt{K}} \int_\Gamma dy\, e^{y^2 g''(y_0)/2} \ .$$

Observe that the contour Γ always crosses the point $y = 0$, and since the integrand is holomorphic on the whole complex plane one can deform Γ so that it is tangent to the real axis at $y = 0$. All points on Γ scale as $\sqrt{K}$ as K grows large, and in the limit $K \to \infty$ the contour Γ simply becomes the real axis with some appendages at infinity. Noting $g''(y_0) = -1$, the eigenvalues can now be written as

$$\lambda_m \simeq \frac{e^{K-m^2/2K}}{2\pi\sqrt{K}} \int_{-\infty}^{\infty} dy\, e^{-y^2/2} = \frac{e^{K-m^2/2K}}{2\pi\sqrt{K}} \left(\sqrt{2\pi}\right) = \frac{1}{\sqrt{2\pi K}} e^{K-m^2/2K} \ .$$

The trick that we've used here is also known as the **Hubbard-Stratonovich transformation**, which has been widely used to calculate the partition function of many-body systems particularly that of interacting quantum systems. For instance, see Phys. Rev. Lett. **3**, 77 (1959) and Phys. Rev. B **40**, 456 (1989).

(c) Recall that the trace of T^N in the thermodynamic limit is dominated by its largest eigenvalue. From part (b), it is easy to see that the eigenvalues are ordered (i.e. $\lambda_0 > \lambda_1 > \cdots$) and the largest eigenvalue λ_0 is non-degenerate. Therefore, in the limit $N \to \infty$ we have

$$\log Z = \log \mathrm{tr}(T^N) \simeq N \log \lambda_0 = NK - \frac{N}{2} \log(2\pi K) \ .$$

The energy per site is therefore

$$E/N = -\frac{1}{N}\frac{\partial \log Z}{\partial \beta} = -\frac{J}{N}\frac{\partial \log Z}{\partial K} = -J + \frac{k_\mathrm{B}T}{2} \ .$$

Here, the first term represents the energy of the perfectly ordered chain and the second term is the energy of harmonic excitations. We have a finite specific heat capacity $c = \partial E/\partial T = Nk_\mathrm{B}/2$ at zero temperature. From the equipartition theorem, this means each site has exactly one degree of freedom

(the angle θ) that can store potential energy. So far we've ignored quantum effects, so there is no quenching of the oscillator energies. At sufficiently low temperatures, quantum effects will invalidate the results obtained here. See Phys. Rev. **127**, 1508 (1962).

(d) Recall that the correlation length is related to the ratio of the two largest eigenvalues, we have

$$\xi = -\frac{1}{\log(\lambda_1/\lambda_0)} = 2K \propto \frac{1}{T} \ .$$

We find that the correlation length scales as T^{-1} at low temperatures. In this case, the divergence of ξ means that the continuous $O(2)$ symmetry of XY model in $d = 1$ is only (spontaneously) broken at $T = 0$. So, there is no phase transition at finite T. This is true for any system with continuous symmetries in $d = 1$ and $d = 2$ as dictated by the Mermin-Wagner theorem, albeit they can still host different type of phase transitions. See Phys. Rev. Lett. **17**, 1133 (1966) and The Nobel Prize in Physics 2016.

Remarks: The mathematical trick we've used relies heavily on $K \to \infty$, thus the form of λ_m we've obtained is only accurate at low temperatures. There is actually an exact solution to one-dimensional XY model, see Phys. Lett. A **104**, 357 (1984). The exact eigenvalues are given by $\lambda_m = I_m(K)$ for all K, where $I_m(K)$ denotes the modified Bessel function of the first kind. As shown in Fig. 1, the approximated eigenvalues converge rather quickly to the exact solution as temperature decreases.

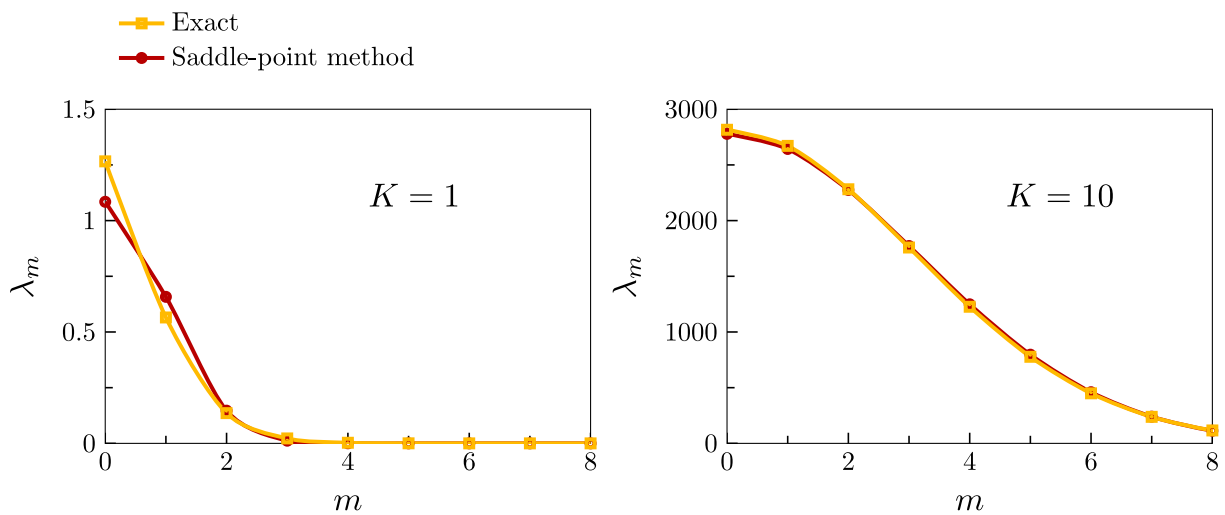

Figure 1: The eigenvalues obtained from the saddle-point method are compared with the exact ones. The approximated eigenvalues are already in excellent agreement with the exact values even at $K = 10$.

4 2D Ising model

In this question, we'll use transfer matrix formalism to the two-dimensional to the Ising model on a square lattice. Suppose that there are N rows parallel to the x-axis and M rows parallel to the y-axis. We will require

that $N \to \infty$ as we calculate the transfer matrix for $M = 1$ and $M = 2$. The Hamiltonian is given by

$$-\beta H = K \sum_{n=1}^{N} \sum_{m=1}^{M} S_{m,n} S_{m+1,n} + S_{m+1,n} S_{m,n+1} \ .$$

We further apply periodic boundary conditions in both directions, i.e. $S_{m,N+1} = S_{m,1}$ and $S_{M+1,n} = S_{1,n}$ so that our lattice has the topology of a torus (see Fig. 2).

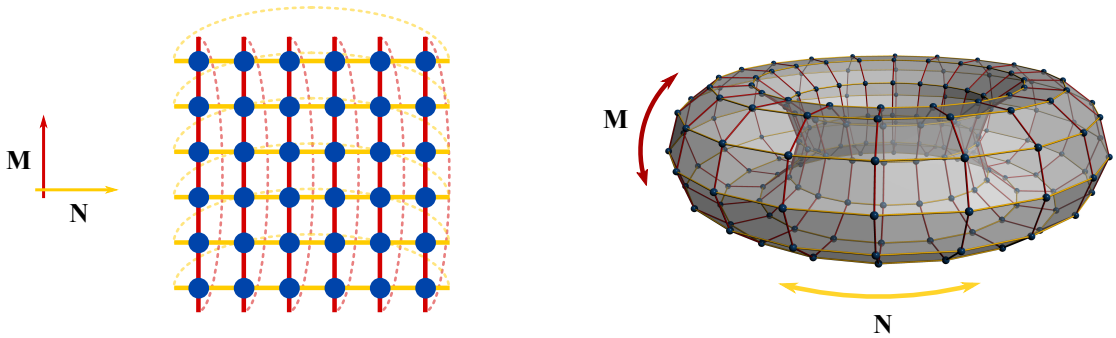

Figure 2: The lattice has boundary conditions in both directions such that its topology is that of a torus.

(a) For $M = 1$, the Hamiltonian reduces to

$$-\beta H = K \sum_{n=1}^{N} (S_{1,n})^2 + S_{1,n} S_{1,n+1} \ .$$

Here, we've used the boundary condition on the M direction, i.e. $S_{M+1,n} = S_{1,n}$. The partition function reads

$$Z = \sum_{\{S\}} \exp(-\beta H) = \sum_{\{S\}} \exp\left[K \sum_{n=1}^{N} 1 + S_{1,n} S_{1,n+1} \right] \ .$$

Suppose we define a 2 by 2 matrix T whose elements are given by

$$T_{n,n'} = \exp\left[K \left(1 + S_{1,n} S_{1,n'} \right) \right] ,$$

the partition function can be written as

$$Z = \sum_{\{S\}} T_{1,2} T_{2,3} \cdots T_{N-1,N} T_{N,1} \ .$$

Now, we can see that T is the transfer matrix. In the basis $\{1, -1\}$, the matrix T reads

$$T = \begin{pmatrix} e^{2k} & 1 \\ 1 & e^{2k} \end{pmatrix} = \begin{pmatrix} x^2 & 1 \\ 1 & x^2 \end{pmatrix} \ ,$$

where we've identified $x = e^K$. Solving the equation $(x^2 - \lambda)^2 - 1 = 0$, one gets $\lambda_1 = x^2 + 1$ and $\lambda_2 = x^2 - 1$.

(b) In this part, we'll first generalize the formalism to M by N lattice. If we now denote a column configuration by $\mathbf{v}_n = (S_{1,n}, S_{2,n}, \ldots, S_{M,n})$, we see that there are a total of 2^M possible configurations for each column and we can write H as a sum of two terms: the interaction energy of columns and the interaction energy between nearest-neighbor columns. We identify

$$-\beta E_1(\mathbf{v}_n, \mathbf{v}_{n+1}) = K \sum_{m=1}^{M} S_{m,n} S_{m,n+1}$$

to be the interaction energy between the n-th and $(n+1)$-th columns, and

$$-\beta E_2(\mathbf{v}_n) = K \sum_{m=1}^{M} S_{m,n} S_{m+1,n}$$

to be the interaction energy of the n-th column. So, we can write

$$H = H\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_N\} = \sum_{n=1}^{N} \left[E_1(\mathbf{v}_n, \mathbf{v}_{n+1}) + E_2(\mathbf{v}_n)\right] \ .$$

The partition function can then be written as

$$Z = \sum_{\{S\}} \exp\left(-\beta H\right) = \sum_{\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_N\}} \exp\left[-\beta \left(\sum_{n=1}^{N} E_1(\mathbf{v}_n, \mathbf{v}_{n+1}) + E_2(\mathbf{v}_n)\right)\right] .$$

Notice that if we define a matrix T whose elements

$$T_{\mathbf{v}_n, \mathbf{v}_{n'}} = \exp\left[-\beta E_1(\mathbf{v}_n, \mathbf{v}_{n'})\right] \exp\left[-\beta E_2(\mathbf{v}_n)\right] = \exp\left[K \sum_{m=1}^{M} S_{m,n} S_{m,n'} + S_{m,n} S_{m+1,n}\right] ,$$

the partition function Z then reads

$$Z = \sum_{\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_N\}} T_{\mathbf{v}_1, \mathbf{v}_2} T_{\mathbf{v}_2, \mathbf{v}_3} \cdots T_{\mathbf{v}_{N-1}, \mathbf{v}_N} T_{\mathbf{v}_N, \mathbf{v}_1} = \sum_{\mathbf{v}_1} (T_{\mathbf{v}_1, \mathbf{v}_1})^N ,$$

where we've used the fact that $\mathbf{v}_{N+1} = \mathbf{v}_1$. It is now clear that the 2^M by 2^M matrix T is our transfer matrix.

(c) Substituting $M = 2$ to the matrix T found in part (b), in the basis $\{(1,1), (1,-1), (-1,1), (-1,-1)\}$ we have

$$T = \begin{pmatrix} x^4 & x^2 & x^2 & 1 \\ x^{-2} & 1 & x^{-4} & x^{-2} \\ x^{-2} & x^{-4} & 1 & x^{-2} \\ 1 & x^2 & x^2 & x^4 \end{pmatrix} .$$

(d) In order to get the eigenvalues easily, it is imperative that we diagonalize T. This can be done as follow

$$\begin{aligned} T_{\mathbf{v}_n,\mathbf{v}_{n'}} &= \exp\left[-\frac{\beta}{2}E_2(\mathbf{v}_{n'})\right]\exp\left[-\beta E_1(\mathbf{v}_n,\mathbf{v}_{n'})\right]\exp\left[-\frac{\beta}{2}E_2(\mathbf{v}_{n'})\right] \\ &= \exp\left[\frac{K}{2}\sum_{m=1}^{M} S_{m,n}S_{m+1,n} + 2S_{m,n}S_{m,n'} + S_{m,n'}S_{m+1,n'}\right] . \end{aligned}$$

For $M = 2$ it is then straightforward to obtain

$$\widetilde{T} = \begin{pmatrix} x^4-1 & 0 & 0 & 0 \\ 0 & 1-x^{-4} & 0 & 0 \\ 0 & 0 & \frac{1}{2}x^4+1+\frac{1}{2}x^{-4}-\frac{1}{2}\sqrt{x^8+x^{-8}+14} & 0 \\ 0 & 0 & 0 & \frac{1}{2}x^4+1+\frac{1}{2}x^{-4}+\frac{1}{2}\sqrt{x^8+x^{-8}+14} \end{pmatrix} ,$$

where $x = e^K$. Since the transfer matrix is now a diagonal matrix, its eigenvalues are simply its diagonal entries. The two largest eigenvalues of T are

$$\lambda_1 = \frac{1}{2}x^4 + 1 + \frac{1}{2}x^{-4} + \frac{1}{2}\sqrt{x^8+x^{-8}+14}, \qquad \text{and} \qquad \lambda_2 = x^4 - 1 .$$

Remarks: The two-dimensional Ising model, whose exact solution was first calculated by Lars Onsager, is historically the first model that was found to exhibit critical behavior at a phase transition, see Phys. Rev. **65**, 117 (1944) and Phys. Rev. **85**, 808 (1952). Much simpler methods to solve the model analytically were developed soon afterwards, see Phys. Rev. **76**, 1244 (1949) and Rev. Mod. Phys. **36**, 856 (1964). Also note that it is important to choose an appropriate boundary condition as it is often the key to make our calculations easier. For instance, in this question we've imposed periodic boundary conditions on both directions. If we only consider a periodic boundary condition on the N direction, the topology of the lattice will be that of a cylinder instead of a torus (see Fig. 3). Obviously, the form of transfer matrix T would also change:

$$T_{\text{cylinder}} = \begin{pmatrix} x^1 & x^{-1} \\ x^{-1} & x^1 \end{pmatrix} \text{ for } M = 1 , \qquad T_{\text{cylinder}} = \begin{pmatrix} x^3 & 1 & 1 & x^{-1} \\ 1 & x & x^{-3} & 1 \\ 1 & x^{-3} & x & 1 \\ x^{-1} & 1 & 1 & x^3 \end{pmatrix} \text{ for } M = 2 .$$

In the thermodynamic limit ($M, N \to \infty$), howerver, we should expect that different boundary conditions give the same physical properties of the model.

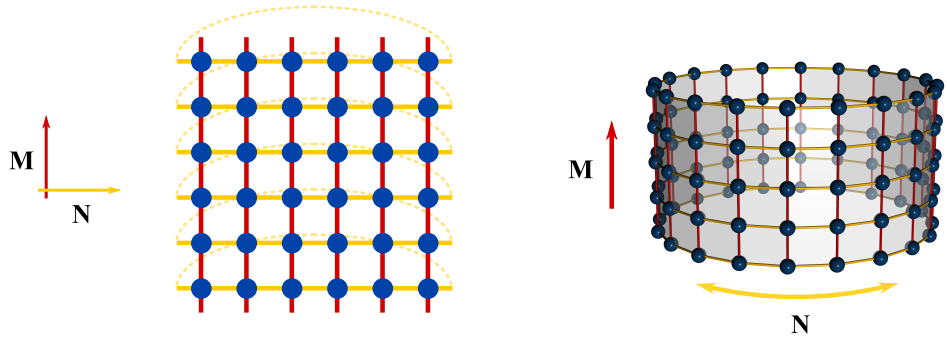

Figure 3: The lattice with a boundary condition only in the N direction, it is now wrapped on a cylinder instead of a torus.